Payman Sharafianardakani

Robotics Research Engineer | Ph.D. Candidate (Expected Dec. 2026) | Full-Stack Roboticist

Louisville, KY • **Work Authorization: EAD Holder (Green Card Pending) – No Sponsorship Required**

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Robotics engineer with 8 years of research and system-integration experience spanning physical human–robot interaction, safety-critical shared control, sensor fusion, and deep learning. Primary engineer on an NSF-funded clinical mobile manipulator, owning the stack end to end: motor-controller PCBs and SLA-printed sensor hardware, ROS control and perception nodes, MPC-CBF safety filters, a browser teleoperation front end, and the IRB human-subject trials that validated all of it. Five peer-reviewed journal papers, two as first author, with two further first-author manuscripts under review and in preparation, plus hospital deployment with real patients and nurses.

Technical Skills

Languages: Python, C++, MATLAB, TypeScript/JavaScript, Bash

Robotics: ROS1 (Kinetic/Melodic/Noetic), ROS2 (Humble), Gazebo, RViz, Motion & Path Planning, MoveIt, ROS Navigation Stack (AMCL, costmaps), TF2, Kinova Kortex, Mecanum omnidirectional kinematics, URDF, NVIDIA Isaac Sim

Control & Optimization: MPC, Control Barrier Functions (CBF-QP, MPC-CBF), OSQP, CasADi, Admittance/Impedance Control, Neuroadaptive Control, Lyapunov Stability Analysis, Genetic Algorithms (PyGAD), Real-Time Control (50–333 Hz loops)

Estimation & Sensor Fusion: Kalman Filtering (Singer motion model), multi-modal tactile + force/torque fusion, LiDAR obstacle sectoring, network-delay estimation

Deep Learning: PyTorch, TensorFlow, Keras, Scikit-learn, ONNX, CUDA, Semantic Segmentation (zero-shot, FastSAM), Temporal Convolutional Networks, BiLSTM, CNN-GRU, MLP, PointNet++, FastSAM, Contact-GraspNet, Reinforcement Learning

Embedded & Systems: Linux, EtherCAT motor-control chains, CAN, SPI, I2C, UART, Arduino, Teensy, Sabertooth motor drivers, Jetson, VersaLogic single-board computers, analog signal conditioning, socket programming

Web & Infrastructure: distributed system architecture, CI/CD (GitHub Actions), Next.js (App Router), React, TypeScript, Tailwind CSS, static-export SPA, WebSockets, rosbridge, Cloudflare Tunnel + Access (SSO), CBOR/JPEG stream compression, Docker, Git

Hardware & Prototyping: SolidWorks (DFM/assembly), FDM and SLA 3D printing, PCB design/debug/soldering, FSR tactile arrays (FlexiForce A201), ATI Axia 80 F/T sensors, multi-channel data acquisition (DAQ), RealSense RGB-D, 3D LiDAR

Research Methods: IRB protocol authoring, within-subjects/counterbalanced experiment design, linear mixed-effects models in R, Technology Acceptance Model (TAM)

Professional Experience

Robotics Research Engineer (Ph.D. Research Assistant) — Sep. 2022 – Present
Louisville Automation & Robotics Research Institute (LARRI), University of Louisville — *Louisville, KY*

- **Primary Engineer, ARNA (NSF Future of Work #FW-HTF-RM)**: Sole lead engineer for the Adaptive Robotic Nursing Assistant, an omnidirectional mobile manipulator (7-DOF Kinova Gen3 on a Mecanum base) deployed in an active hospital. Owned hardware, control, perception, web infrastructure, and clinical trials.
- **Network-Aware Safety Architecture for Remote Teleoperation** (first author, in preparation, *IEEE T-RO*): Architected a five-layer shared-control stack that holds hard safety guarantees when *both* the operator and the connection are unreliable, under one invariant: adaptive layers may only make the robot more cautious, and barrier constraints are never relaxed. A 100 Hz arm MPC-CBF gave **zero boundary violations in 60 runs** at jerk cost **200–440$\times$ lower** than raw operator input; a 50 Hz OSQP base filter on LiDAR sectors gave zero margin violations; a watchdog degraded margins in $\sim$100 ms against a 200 ms requirement.
- **Tactile Handlebar & Real-Time Adverse-Event Detection** (first author, under review, *IEEE T-MRB*): Led end to end – mechanical design, electronics, ML, and human trials. Designed a sensorized handlebar with **eight FlexiForce A201 FSRs** and a custom Arduino signal-conditioning board driving **synchronous 8-channel data acquisition** at **122 Hz**, fused with 6-axis force/torque through a **Singer-model Kalman filter**. Benchmarked TCN vs. BiLSTM vs. CNN-GRU under subject-wise cross-validation, then deployed the TCN to ONNX onboard the robot – **no cloud** – re-evaluating every **8.2 ms**. On **10 unseen participants**: **85.0% detection** at **2.05 s mean latency**.
- **Adaptive Teleoperation Interface (PNNUI)** (first author, *IEEE RA-L* 2025): Learned the joystick-to-robot mapping as a black box, with no kinematic model of either device: two networks in parallel, one searched offline by a genetic algorithm to minimize task time, one trained online by SGD against measured jerk. In a 20-subject IRB study, **linear jerk halved** and **angular jerk fell $\sim$3$\times$** ($p < .001$), **with no increase in task-completion time**.
- **Neuroadaptive Admittance Control** (first author, *Intelligent Service Robotics* 2025): Designed a model-free inner-loop controller that learns ARNA's unmodelled dynamics online, with update laws derived from a Lyapunov analysis, running at 333 Hz against a 125 Hz admittance model. Against a tuned PD baseline with 10 users: tracking accuracy improved up to **32%** and motion jerk fell up to **52%**. Re-tuned with **63 nursing students**: velocity tracking error **0.0157 $\rightarrow$ 0.0105 m/s**, mean actuator torque **3.66 $\rightarrow$ 1.79 Nm**.

- **Browser-Based Teleoperation Front End & Secure Transport**: Built a **TypeScript**/Next.js interface an operator opens as a URL – no client install, no VPN, and **no inbound port on the robot**, via an outbound Cloudflare Tunnel with identity-based Access. Compiled to a **fully static bundle** with **no Node runtime on the robot**, so all robot communication is client-side over WebSocket. Split the session across **three isolated rosbridge instances** so camera bursts cannot head-of-line-block a joystick or stop command, and cut per-frame data volume $\sim$**35–40**$\times$ with source-side JPEG encoding and CBOR subscriptions. Evaluated with **30 remote operators** across multiple U.S. cities, so the latency the safety layers compensate for is measured rather than simulated.
- **One-Click Semi-Autonomous Grasping**: Reduced remote manipulation to a single click on a single pixel: FastSAM segments the clicked object on CPU, and Contact-GraspNet with a PointNet++ encoder proposes ranked SE(3) grasps. Deployed on a **4 GB laptop GPU** by cropping inference to the object bounding region ($\sim$10$\times$ input reduction) and warm-loading weights once at node start. Re-ranked candidates **by reachability rather than network confidence** – the binding constraint in practice.
- **Hospital Clinical Trial** (first author, *IEEE CASE* 2025): Designed and led an acceptability trial on an active unit at University of Louisville Hospital with **10 patients and 5 nurses** – real patients, not a lab surrogate. Paired objective task-time logging with Technology Acceptance Model instruments, analyzed with linear mixed-effects models in R: the joystick was significantly easier to use and more likely to be adopted than the tablet ($p \approx 0.02$–0.03) *despite statistically similar task times*.
- **Hardware Recovery & Systems Integration**: Repaired and upgraded ARNA's motor-controller PCBs, brought non-functional sonar and bump sensors back online and into ROS, repurposed the LiDAR for navigation, and rebuilt onboard wiring across the EtherCAT motor chain, Sabertooth motor drivers, Teensy microcontrollers, battery/inverter bank, and instrumentation board. Worked across the robot's serial and fieldbus layers – **EtherCAT, CAN, SPI, I2C, and UART** – to bring sensors and drives back under ROS control.
- **Mentorship & Documentation**: Mentored **one M.S. student and six undergraduate researchers**, and authored the lab manuals and Standard Operating Procedures for ARNA that new researchers onboard from.

Mechatronics Engineer (M.Sc. Research Assistant) Nov. 2017 – Sep. 2021
University of Tehran – Human-Robot Interaction Lab *Tehran, Iran*

- **Audio-Visual Perception System**: Engineered a sound-source localization algorithm for an interactive social robot (autism therapy). Achieved a **70% improvement** in audio filtering and **40% increase** in gesture detection accuracy.
- **Embedded Mobile Robotics**: Designed the "Baby Tank" open-source platform. Developed navigation/path-planning algorithms and managed low-level microcontroller integration for active damping.
- **Application Development**: Built a cross-platform Activity Reminder GUI using **C++ and Qt Creator**, focusing on accessibility standards.

Education

Ph.D. Candidate, Electrical & Robotics Engineering, University of Louisville, KY Sep. 2022 – **Dec. 2026 (expected)**
M.S. in Mechatronics Engineering, University of Tehran, Tehran, Iran Nov. 2017 – Sep. 2021
B.Sc. in Electrical Power Engineering, Islamic Azad University, Varamin, Iran Sep. 2012 – Aug. 2017

Selected Publications

Full list: paymansharafian.github.io/publications • Google Scholar

- **P. Sharafianardakani**, M. A. Hanafy, I. Kondaurova, A. Ashary, M. M. Rayguru, D. O. Popa. "Adaptive User Interface With Parallel Neural Networks for Robot Teleoperation." *IEEE Robotics and Automation Letters*, 10(2):963–970, 2025.
- **P. Sharafianardakani**, S. Abubakar, S. K. Das, S. Surupa, M. A. Hanafy, M. Rayguru, D. O. Popa. "Evaluation of a Neuroadaptive Admittance Controller for Ambulation through Physical Human–Robot Interaction." *Intelligent Service Robotics*, 18(7):1355–1368, 2025.
- **P. SharafianArdakani**, I. Kondaurova, A. Ashary, N. Zhang, M. C. Logsdon, M. Walker, D. O. Popa. "Hospital Trial Results for the Acceptability of an Adaptive Robot Nursing Assistant." *IEEE CASE*, pp. 2043–2049, 2025.
- I. Kondaurova, **P. Sharafian**, R. Mitra, M. M. Rayguru, *et al.*, D. O. Popa. "Acceptance of an Adaptive Robotic Nursing Assistant for Ambulation Tasks." *Robotics* (MDPI), 14(9):121, 2025.
- M. C. Logsdon, I. Kondaurova, N. Zhang, *et al.* (incl. **P. SharafianArdakani**), D. O. Popa. "Perceived Usefulness of Robotic Technology for Patient Fall Prevention." *Workplace Health & Safety*, 72(12):542–549, 2024.
- **P. Sharafianardakani** *et al.* "Tactile Handlebar and Deep Learning for Safe Interaction with a Robot Nursing Assistant." *IEEE Transactions on Medical Robotics and Bionics* – **under review**.

Professional Service & Honors

Journal Reviewer, IEEE Robotics and Automation Letters (RA-L), 2024–2025 • **Conference Reviewer**, ACM/SIGAPP Symposium on Applied Computing (SAC), 2024

STEM Mentor, RAMPS Program – 30+ high-school and undergraduate students, 2023–Present

Dissertation Completion Award (full tuition + stipend), University of Louisville, 2026

IEEE RAS Travel Support Award (ICRA 2025), IEEE Robotics and Automation Society • **Graduate Student Travel Award**, University of Louisville, 2025